

Day 22 Hybrid Notes

How changing size changes measurements | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can explain how scale factor affects length, area, and volume.
I can use density as mass divided by volume.
I can decide whether a design meets a given constraint.

Part 1: Cloze Notes - Essential Ideas
1. If lengths scale by k, perimeters and lengths scale by _____.
2. If lengths scale by k, areas scale by _____.
3. If lengths scale by k, volumes scale by _____.
4. Density equals mass divided by _____.
5. Surface area is measured in _____.
6. Volume is measured in _____.
7. A design constraint is a limit or requirement a solution must _____.

Part 2: Outline Notes - Big Picture A. Similarity connects scale factor to _____. B. Area scale is connected to two dimensions: length and _____. C. Volume scale is connected to three dimensions: length, width, and _____. D. Modeling problems require choosing formulas and checking _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

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Guided setup + active learning

Part 3: Guided Setup Practice

1. Two similar solids have scale factor 3. The smaller volume is 40 cubic units. Find the larger volume.

Setup/work: _____

Answer: _____

2. A solid has mass 500 g and volume 125 cm^3 . Find density.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student says a scale factor of 2 means area and volume both double.

What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A package can weigh no more than 2000 g. A solid object has volume 300 cm^3 and density 8 g/cm^3 . Does it meet the constraint?

Setup: _____

Answer: _____

Sentence answer: _____